Science practice booklet: Grade 10 expectations

Grade 10 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 10 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: HS-LS2-1, HS-LS2-2, HS-LS2-4, HS-LS2-6, HS-LS3-1, HS-LS3-2, HS-LS3-3, HS-LS4-3, HS-LS4-5

A worked example

In a mangrove ecosystem, algae capture 20000 units of energy from sunlight. Small fish eat the algae, crabs eat the small fish, and herons eat the crabs. About 10 percent of the energy is passed on at each step. How much energy reaches the herons?

1. Algae are the producers. Small fish are the first consumers, crabs the second consumers and herons the third consumers.
2. Algae to small fish: 10 percent of 20000 = 2000 units.
3. Small fish to crabs: 10 percent of 2000 = 200 units.
4. Crabs to herons: 10 percent of 200 = 20 units.
5. The rest of the energy at each step is used for life processes or given off as heat.

About 20 units of energy reach the herons.

Questions

1. What is meant by the carrying capacity of an ecosystem?

   A. The number of species living in the ecosystem

   B. The number of young born in one year

   C. The largest population of one species the ecosystem can support over time

   D. The area of the ecosystem in square kilometres

2. A reserve starts with 100 Arabian oryx. The numbers rise quickly for several years, then stay close to 600 for a long time. What does the number 600 show?

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3. Name two limiting factors that can keep a desert animal population from growing, and say which are living (biotic) and which are non-living (abiotic).

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4. Producers in a coastal lagoon capture 50000 units of energy. About 10 percent is passed on at each step. How much energy reaches the second consumers?

   A. 5000 units

   B. 500 units

   C. 50 units

   D. 45000 units

5. On a date farm the palms grow taller each year. Describe how carbon moves between the air and the palms and back again. Name the two key processes.

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6. Why does a mangrove forest with many different species usually recover better after a storm than a place with only one or two species?

   A. Forests with many species always have taller trees

   B. A place with one species has more energy to spend

   C. Storms only harm places that have many species

   D. More species give more food sources and links, so others can keep the ecosystem working

7. Red flowers (R) are dominant and white flowers (r) are recessive in a garden plant. Two plants with the genotype Rr are crossed. What fraction of the offspring is expected to have white flowers?

   A. One quarter

   B. None

   C. One half

   D. Three quarters

8. At a falcon breeding centre, chicks from the same two parents differ in feather colour and size. Give two ways in which variation between individuals of the same species can arise.

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9. Some desert plants have thick, waxy stems and deep roots. Explain how these traits help the plants survive.

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10. Arabian oryx have pale coats that reflect sunlight and hooves that spread weight on soft sand. Explain how natural selection could produce these traits in a group over many generations.

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11. Why can a population of animals change over many generations when one individual cannot change its inherited traits?

    A. An individual can change its genes by trying hard

    B. Traits gained by exercise are passed on to the young

    C. Helpful traits become more common because individuals that have them leave more offspring

    D. Every individual changes its genes a little each day

12. Building along a coast removes mangrove trees, and the number of fish and crabs there falls. Explain two ways in which human activity can reduce biodiversity, and suggest one action that helps.

Section 2: Earth and space

Standards practised: HS-ESS1-1, HS-ESS1-3, HS-ESS1-4, HS-PS2-4

A worked example

Light travels at about 300 000 km per second. Mars is on average about 228 000 000 km from the Sun. How long does sunlight take to reach Mars?

1. Time = distance ÷ speed.
2. 228 000 000 ÷ 300 000 = 760 seconds.
3. 760 ÷ 60 = 12 with 40 left over, so it is 12 minutes 40 seconds.

About 760 seconds, which is about 12 minutes 40 seconds.

Questions

1. Which process produces the energy given out by the Sun?

   A. Burning hydrogen gas with oxygen

   B. Nuclear fusion of hydrogen nuclei into helium

   C. Nuclear fission of heavy atoms

   D. Reflecting light from other stars

2. The Sun stays about the same size for billions of years. Which two things are in balance inside it?

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3. Which list shows the stages of a star like the Sun in the correct order?

   A. Nebula, red giant, main sequence star, white dwarf

   B. Main sequence star, nebula, red giant, white dwarf

   C. Nebula, main sequence star, white dwarf, red giant

   D. Nebula, main sequence star, red giant, white dwarf

4. How does the end of a star with a very large mass differ from the end of the Sun?

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5. Elements such as carbon and oxygen are made inside stars. How can this matter later become part of new planets?

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6. The Hope Probe sends a radio signal to Earth when Mars and Earth are 60 000 000 km apart. Radio waves travel at the speed of light, 300 000 km per second. How long does the signal take?

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7. What does a light year measure?

   A. The distance light travels in one year

   B. A length of time equal to one year

   C. The brightness of a star over one year

   D. The time light takes to cross the Solar System

8. Scientists split starlight into a spectrum and see dark lines at certain colours. What do the lines show?

   A. How fast the star is spinning

   B. How old the star is

   C. Which elements are in the star's atmosphere

   D. How far away the star is

9. From a desert camp, Hamad sees two stars that are the same distance away. One looks blue-white and the other looks red. Which one has the hotter surface?

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10. A satellite moves to twice its distance from the centre of Earth. How does the pull of Earth on it change?

    A. It becomes half as large

    B. It becomes one quarter as large

    C. It becomes twice as large

    D. It stays the same

11. Planets closer to the Sun complete an orbit in less time than planets farther away. Mercury takes 88 days and Earth takes 365 days. Explain why this is expected.

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12. The Hope Probe travels in an elliptical (oval) orbit around Mars. Using gravity, explain why the probe moves fastest when it is nearest to Mars.

Section 3: Matter, forces and energy

Standards practised: HS-PS2-1, HS-PS2-2, HS-PS2-3, HS-PS2-4

A worked example

A trolley of mass 1000 kg moves at 20 m per second and joins with a parked trolley of mass 1000 kg so that they move together. What is their speed after joining? (Ignore friction.)

1. Momentum = mass × velocity.
2. Momentum before = 1000 × 20 + 1000 × 0 = 20000 kg m/s.
3. Momentum is conserved, so after joining it is still 20000 kg m/s.
4. Total mass = 2000 kg, so velocity = 20000 ÷ 2000 = 10 m per second.

10 m per second.

Questions

1. A bus brakes suddenly and the standing passengers lurch forward. Which idea explains this?

   A. Their bodies tend to keep moving at the same speed because no force stops them

   B. The brakes push the passengers forward

   C. A force is always needed to keep an object moving

   D. Passengers become lighter when the bus slows

2. A scooter and its rider have a total mass of 80 kg. An unbalanced force of 240 N pushes them forward. What is their acceleration? Use force = mass × acceleration.

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3. The same force pushes a 2 kg trolley and a 4 kg trolley. Which statement is correct?

   A. The 4 kg trolley accelerates twice as much

   B. Both trolleys accelerate equally

   C. The 2 kg trolley accelerates half as much

   D. The 2 kg trolley accelerates twice as much

4. What is the difference between mass and weight? Find the weight of a 50 kg astronaut where the gravitational field strength is 3.7 N per kg (as on Mars). Use weight = mass × field strength.

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5. A rocket pushes hot gas out of its engine downwards. Why does the rocket move upwards?

   A. The exhaust gas pushes against the air around the rocket

   B. Gravity pulls the rocket upwards

   C. The rocket pushes the gas down, so the gas pushes the rocket up with an equal force

   D. The rocket is lighter than the air

6. A delivery van of mass 1500 kg travels along Sheikh Zayed Road at 20 m per second. What is its momentum? Use momentum = mass × velocity.

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7. A cart of mass 2 kg moves at 3 m per second and hits a cart of mass 1 kg that is at rest. They stick together. What is their speed afterwards?

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8. A football of mass 0.5 kg is kicked from rest and leaves the foot at 20 m per second. The foot is in contact with the ball for 0.02 seconds. Use force = change in momentum ÷ time to find the average force.

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9. Why does an airbag reduce the force on a passenger in a car crash?

   A. It makes the change in momentum smaller

   B. It makes the stopping time longer, so the average force is smaller

   C. It makes the stopping time shorter

   D. It adds extra mass to the passenger

10. Salem drops a stone into a deep well. It falls for 3 seconds before it hits the water. Ignore air resistance and use g = 10 m per second squared. Find its speed when it hits the water and the distance it has fallen.

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11. Two objects attract each other with a gravitational force. What happens to the force if the mass of one object is doubled and the distance stays the same?

    A. It doubles

    B. It stays the same

    C. It becomes four times as large

    D. It halves

12. Two small balloons are rubbed with cloth so that both gain the same kind of charge. What happens when they are brought near each other? By what factor does the electric force change if the distance between two charges is halved?

Answer guide for parents

Section 1: Living things

1. C. The largest population of one species the ecosystem can support over time. Carrying capacity is the largest population of one species that food, water and space can support over time. If your child chose A, they counted species, but it is about the number of individuals of one species.
2. The numbers level off near 600, so food, water and space can support only about 600 oryx. That limit is the carrying capacity of the reserve.
3. Food supply and predators are living (biotic) limits, because they come from living things. Water supply and high temperature are non-living (abiotic) limits.
4. B. 500 units. Take 10 percent at each step: 50000 × 0.1 = 5000, then 5000 × 0.1 = 500. If your child chose A, they stopped after one step, which only reaches the first consumers.
5. Palms take carbon dioxide from the air and build it into sugars by photosynthesis. Animals eat plants, and plants, animals and decomposers return carbon dioxide to the air by respiration.
6. D. More species give more food sources and links, so others can keep the ecosystem working. More species means more food sources and links, so if one is harmed others keep the ecosystem working. If your child chose A, they mixed up variety with height, but tree size does not decide recovery.
7. A. One quarter. The offspring are RR, Rr, Rr and rr, and only rr is white, so 1 out of 4. If your child chose D, they counted Rr as white, but R is dominant, so Rr is red.
8. Mutations change the DNA, and sexual reproduction mixes genes from two parents into new combinations. That is why chicks from the same parents look different.
9. The waxy stem loses less water by evaporation, and deep roots reach water far below the surface. Plants with these traits survive and reproduce more often.
10. Oryx vary, and those with pale coats and wide hooves survived heat and sand better, so they had more young. The traits are inherited, so over many generations they became common.
11. C. Helpful traits become more common because individuals that have them leave more offspring. One animal's genes stay the same, but animals with helpful traits leave more offspring, so the trait spreads. If your child chose B, they thought exercise changes the genes passed on, and it does not.
12. Habitat loss removes shelter and nurseries, and pollution or over-use of resources reduces the number of species. Protecting and replanting mangroves, or making protected reserves, helps.

Section 2: Earth and space

1. B. Nuclear fusion of hydrogen nuclei into helium. In the Sun's core hydrogen nuclei join to make helium, and a little mass becomes a lot of energy. If your child chose A, they thought the Sun burns like a fire, but the Sun runs on fusion, not burning.
2. Gravity pulls inward, while the pressure of the hot gas pushes outward. When these two balance, the Sun stays the same size.

3. D. Nebula, main sequence star, red giant, white dwarf. The Sun's order is nebula, main sequence, red giant, white dwarf. If your child chose A, they put the red giant before the main sequence, but a star lives most of its life as a main sequence star first.
4. A very large star ends in a supernova explosion and leaves a neutron star or black hole. The Sun has less mass, so it ends quietly as a white dwarf.
5. Large stars explode, or older stars throw off their outer layers, and spread carbon and oxygen into space. Gas and dust later gather into new stars and planets.
6. Time = distance ÷ speed = 60 000 000 ÷ 300 000 = 200 seconds. 200 seconds is 3 minutes 20 seconds.
7. A. The distance light travels in one year. A light year is a distance: how far light travels in one year, about 9.5 trillion km. If your child chose B, the word year made them think it is a time, but it measures distance.
8. C. Which elements are in the star's atmosphere. Each element absorbs its own colours, so the pattern of dark lines is like a fingerprint of the elements. If your child chose D, they thought the lines show distance, but they show what the star is made of.
9. The blue-white star. Hotter objects give out light of shorter wavelength, which looks bluer, so the red star is cooler.
10. B. It becomes one quarter as large. Force depends on 1 ÷ distance squared, and doubling the distance gives 2 × 2 = 4, so the pull is one quarter. If your child chose A, they halved it and forgot to square the distance.
11. Gravity from the Sun is stronger on nearer planets, so they move faster. They also have a shorter path, so they finish an orbit sooner.
12. Gravity is stronger when the probe is nearer, so it speeds up as it falls towards Mars. Farther away the pull is weaker, so it moves more slowly.

Section 3: Matter, forces and energy

1. A. Their bodies tend to keep moving at the same speed because no force stops them. Newton's first law: with no unbalanced force on their bodies, they keep moving at the speed the bus had. If your child chose B, they thought the brakes push the passengers, but the brakes only act on the bus.
2. Acceleration = force ÷ mass = 240 ÷ 80 = 3 m per second squared.
3. D. The 2 kg trolley accelerates twice as much. Acceleration = force ÷ mass, so half the mass gives double the acceleration. If your child chose A, they thought the heavier trolley speeds up more, but more mass means less acceleration.
4. Mass is the amount of matter, in kg, and never changes. Weight is gravity's pull in newtons, so weight = 50 × 3.7 = 185 N.
5. C. The rocket pushes the gas down, so the gas pushes the rocket up with an equal force. The rocket pushes the gas down, and the gas pushes the rocket up with an equal force. If your child chose A, they thought the gas pushes on the air, but rockets also work in space where there is no air.
6. Momentum = mass × velocity = 1500 × 20 = 30000 kg m/s.
7. Momentum before is 2 × 3 = 6 kg m/s, and it is kept. The total mass is now 3 kg, so speed = 6 ÷ 3 = 2 m per second.
8. Change in momentum = 0.5 × 20 = 10 kg m/s. Force = 10 ÷ 0.02 = 500 N.

9. B. It makes the stopping time longer, so the average force is smaller. The change in momentum is the same, so a longer stopping time means a smaller average force. If your child chose C, they picked a shorter time, which would make the force bigger.
10. Speed = g × time = 10 × 3 = 30 m per second. Distance = ½ × 10 × 3 × 3 = 45 m.
11. A. It doubles. Force is proportional to the masses, so doubling one mass doubles the force. If your child chose C, they squared the mass, but only the distance is squared, not the mass.
12. Both balloons have the same charge, and like charges repel. Force depends on 1 ÷ distance squared, so halving the distance makes it 2 × 2 = 4 times as large.